Text Retrieval and Search Engines

Lecture 10 – Relevance Models

Some slides adapted from the companion website to the "Introduction to IR" book by Manning, Raghavan and Schutz & Victor Lavrenko

Semester 1 - 2026

Motivation

- PRP: $p(R|D,q) \cong \frac{P(R|D,q)}{P(NR|D,q)} = \prod_{w\in D}(\frac{p_w(1-r_w)}{r_w(1-p_w)})$

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relevance model

background model

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- Background model -> entire set of documents in the collection
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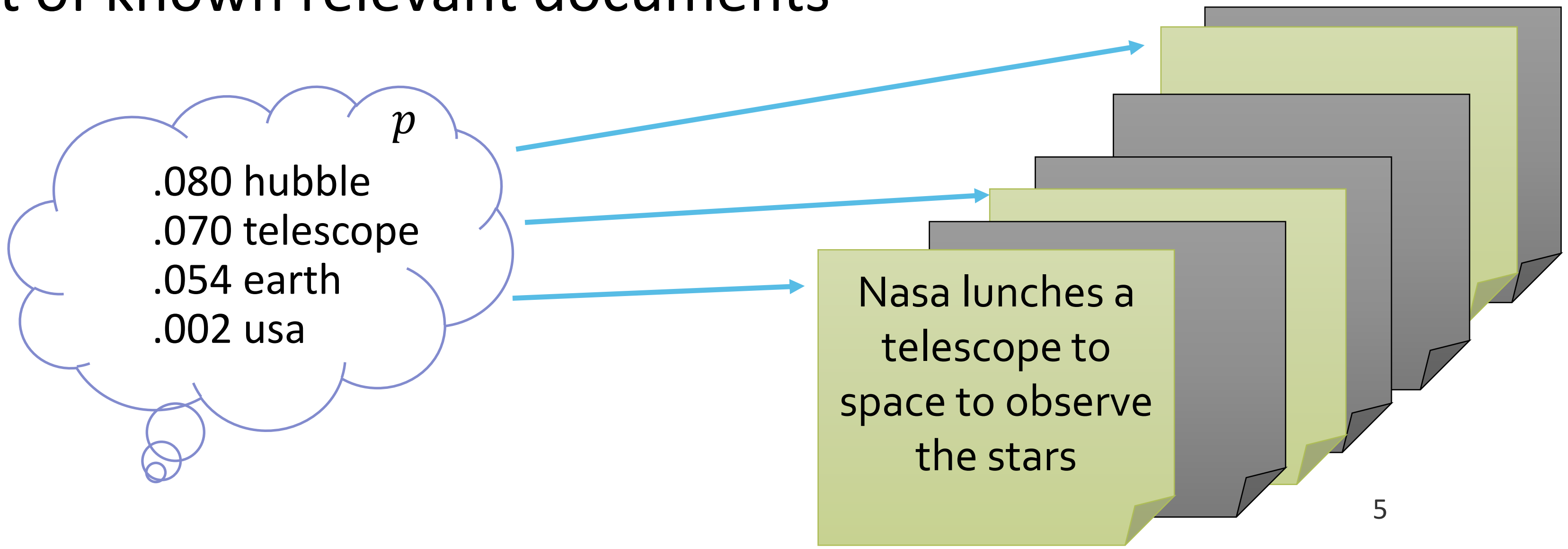

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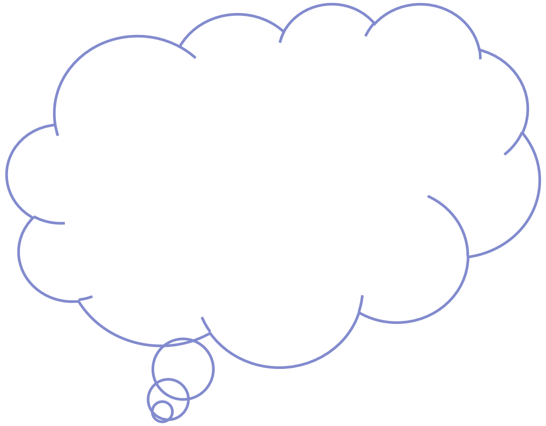

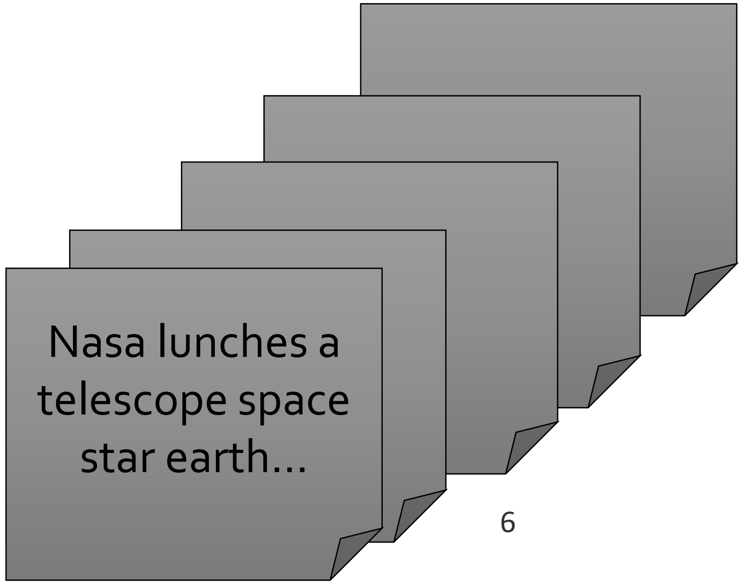

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- Background model -> entire set of documents in the collection
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  - What if we have no relevant examples?
  - Assume query q is a random sample from p

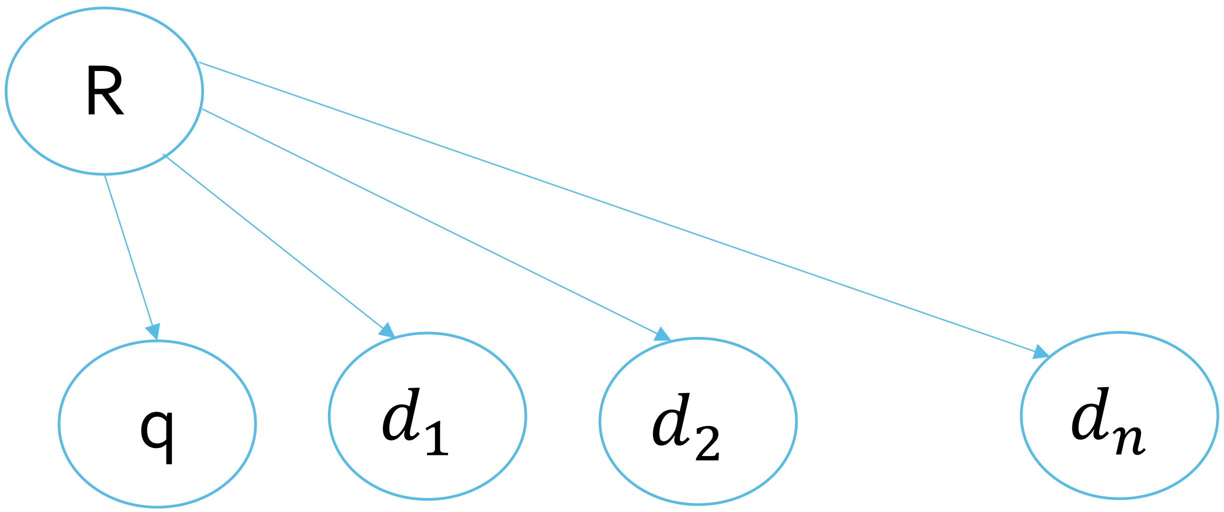
Query: “hubble telescope achievements”

.080 hubble
.070 telescope
.054 earth
.002 usa

Nasa lunches a telescope space star earth…

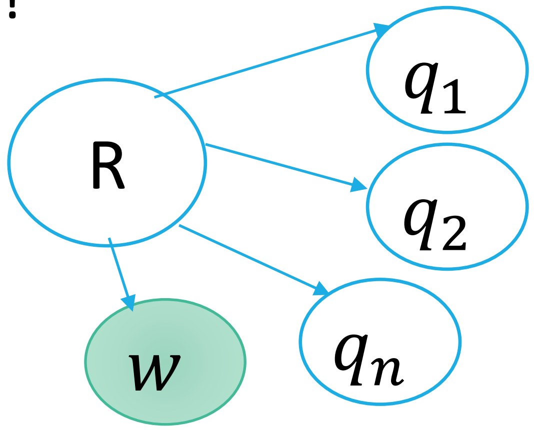
Estimation: The Sampling Game

- Given query q and relevant documents $d_1, d_2,\ldots, d_n$ there is a relevance language model R that generates the terms both in q and in $d_1, d_2,\ldots, d_n$



Estimation: The Sampling Game

- The query and the documents are samples from an unknown relevant model R
- The sampling game
  - Relevant examples are missing
  - Let $q = q_1 \dots q_n$ - the only sample that we have from the R distribution
  - Unknown distribution P, sample 3 times, get: **hubble**, **telescope**, **achievements**
  - What's the chance we get a word **earth** if we sample one more time?



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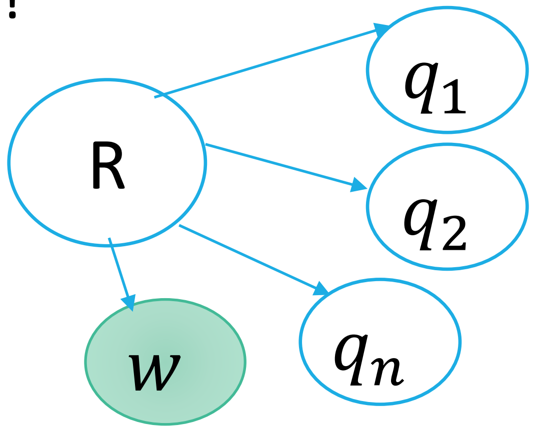
  - $P(earth|R) \approx p(earth|hubble, telescope, achievements)$



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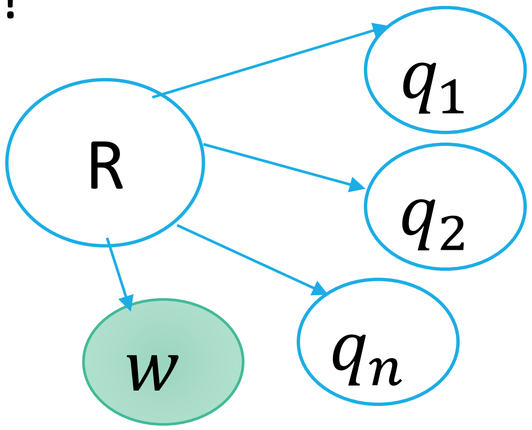


  - $P(earth|R) \approx p(earth|hubble, telescope, achievements) = \dfrac{p(earth, hubble, telescope, achievements)}{p(hubble, telescope, achievements)}$

Estimating the Joint Distribution

- $p(w, q_1 \dots q_n) = ?$

Estimating the Joint Distribution

- $p(w, q_1 \ldots q_n) = ?$
- Assume that the query words $q_1 \ldots q_n$ and the words w in relevant documents are sampled identically and independently from a unigram distribution M_R
- Let M represent some finite universe of unigram distributions from which we could sample
- Sampling process: pick one distribution $M \in$ M with probability $p(M)$ and sample from it n+1 times
- $p(w, q_1 \ldots q_n) = \sum_{M \in \mathrm{M}} p(M) p(w, q_1 \ldots q_n | M)$

I.I.D Sampling

- $p(w, q_1 \dots q_n) = \sum_{M \in \mathrm{M}} p(M) p(w, q_1 \dots q_n | M)$
- Using the I.I.D assumption we get that the conditional probability is

$$p(w, q_1 \dots q_n | M) = p(w|M) \prod_{i=1}^{n} p(q_i | M)$$

- Now we can estimate the joint probability
- $p(w, q_1 \dots q_n) = \sum_{M \in \mathrm{M}} p(M) p(w|M) \prod_{i=1}^{n} p(q_i|M)$

Shortcut for the Estimation

- $p(w, q_1 \ldots q_n) = \sum_{d \in D_{init}} p(M_d) p(w|M_d) \prod_{i=1}^{n} p(q_i|M_d) = \sum_{d \in D_{init}} p(M_d) p(w|M_d) p(q|M_d) = \sum_{d \in D_{init}} p(w|M_d) p(M_d|q) p(q)$
- Recall that $p(w|R) \approx \frac{p(w, q_1, \ldots, q_n)}{p(q)}$
- Therefore, $p(w|R) \approx \sum_{d \in D_{init}} p(w|M_d) p(M_d|q)$
- $p(M_d|q) = \frac{p(q|M_d) p(M_d)}{\sum_{d \in D_{init}} p(M_d) p(q|M_d)}$

Summary

- Basic relevance model: (RM1)
- $p(w|R) \approx \sum_{d \,\in\, D_{init}} p(w|M_d)p(M_d|q)$
- $p(M_d|q) = \frac{p(q|M_d)p(M_d)}{\sum_{d \in D_{init}} p(M_d)p(q|M_d)}$

- Interpolated relevance model (RM3, Abdul Jaleel et al. '04)
- $p(w|R) \approx \beta p_{MLE}(w|q) + (1-\beta) \sum_{d \,\in\, D_{init}} p(w|M_d)p(M_d|q)$

Ranking

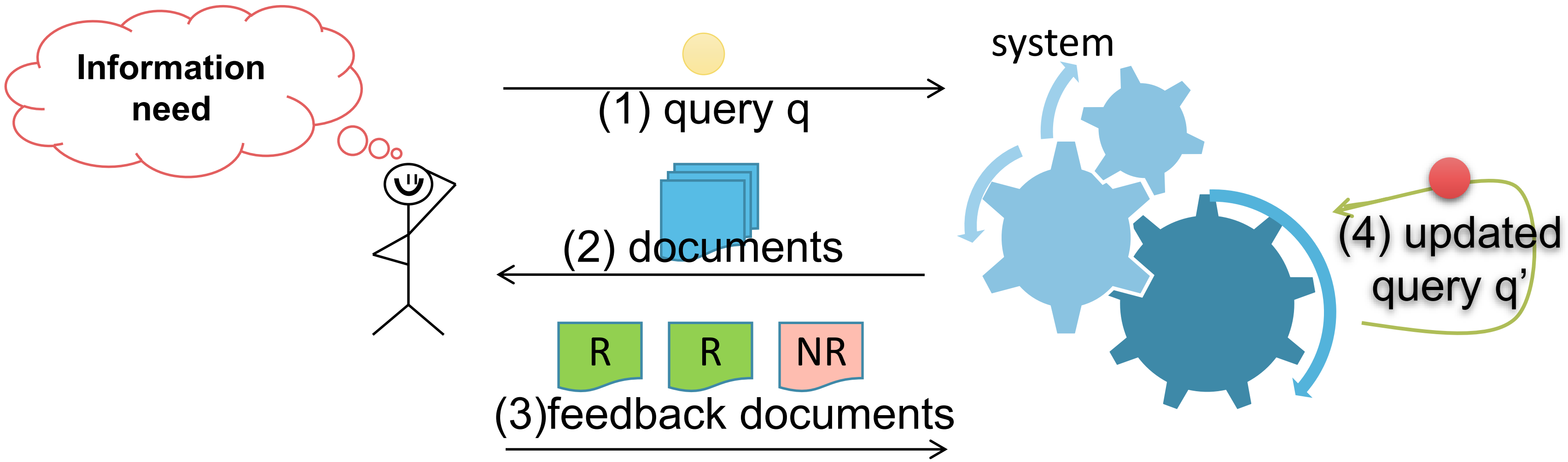
- Compute the KL divergence between a document language model and R
- $Score(d) = -D(p(\cdot\,|R)||p(\cdot\,|M_d))$
- $Score(d) = -\sum_w p(w|R) ln \frac{p(w|R)}{p(w|M_d)}$

Empirical Performance

	ROBUST			WT10G		
	MAP	p@5	NDCG@5	MAP	p@5	NDCG@5
Initial	24.8	47.9	50.0	19.6	33.4	32.8
RM1 (OptAQ)	23.4	48.6	50.9	16.6^{i}	33.6	31.2
RM3 (OptAQ)	28.1^{i}_{r}	49.8	51.9	21.7^{i}_{r}	34.2	33.0
RM1 (OptPQ)	28.7^{i}	56.5^{i}	59.5^{i}	23.2^{i}	43.1^{i}	41.4^{i}
RM3 (OptPQ)	$\mathbf{34.4}^{i}_{r}$	$\mathbf{59.6}^{i}_{r}$	$\mathbf{62.5}^{i}_{r}$	$\mathbf{30.3}^{i}_{r}$	$\mathbf{49.1}^{i}_{r}$	$\mathbf{47.6}^{i}_{r}$
RM1 (LOO)	23.4	48.6	50.9	16.6^{i}	33.6	31.2
RM3 (LOO)	27.3^{i}_{r}	48.5	50.5	21.2_{r}	33.8	32.2
RM1 (10Fold)	23.4	48.6	50.9	15.7^{i}	32.0	29.4
RM3 (10Fold)	27.9^{i}_{r}	49.2	51.3	20.4_{r}	34.0	32.1

Table 1: OptAQ– optimizing MAP over all queries, OptPQ– optimizing MAP per query, LOO– leave-one-out using MAP for optimization, 10Fold– 10-fold cross validation. i,r - statistically significant differences with Initial and RM1.

Relevance Feedback



- Allows the system to update the original query representation
  - May be iterative
  - May be implicit (e.g., from clicks)
- Pseudo-relevance feedback:
  - No judgments given, but we assume relevance of the first documents retrieved

Relevance Feedback

- Pseudo feedback
  - Retrieve an initial set of n documents
  - Assume that the top m ranked documents are relevant
  - Induce query model
    - $P(M_d|q) = \frac{P(q|M_d)P(M_d)}{\sum_{d \in D_{init}} P(M_d)P(q|M_d)}$

	Rank	Doc
"Relevant"	1	d1
"Relevant"	2	d2
"Relevant"	3	d3
	4	d4
	5	d5
	6	d6

- True feedback
  - Retrieve an initial set of n documents
  - The user marks m relevant documents
    - Top-down process
  - Induce query model
    - $P(M_d|q) = \frac{1}{m}$

	Rank	Doc
Relevant	1	d1
Relevant	2	d2
	3	d3
	4	d4
Relevant	5	d5
	6	d6

Example

- Rank the following 4 documents in response to the query “orange apple”
  - d1: orange orange lemon lemon clementine apple
  - d2: orange and red are beautiful colors
  - d3: orange and cellcom offer a special deal
  - d4: the fruits I like most are orange and apple
  - Construct a relevance model using the top 2 documents

Relevance Model Example

- Rank the following 4 documents in response to the query “orange apple”
  - We assume: Dirichlet smoothing with μ=1000

$$p_{Dir}(w \mid M_d) = \frac{tf(w \in d) + \mu p_{MLE}(w \mid M_C)}{dl_d + \mu}$$

- $p(q|d1) = p(orange|d1) * p(apple|d1) = \left(\frac{2+\frac{5}{28}1000}{6+1000}\right) * \left(\frac{1+\frac{2}{28}1000}{6+1000}\right) = 0.0129$
- $p(q|d2) = p(orange|d2) * p(apple|d2) = \left(\frac{1+\frac{5}{28}1000}{6+1000}\right) * \left(\frac{0+\frac{2}{28}1000}{6+1000}\right) = 0.01267$
- $p(q|d3) = p(orange|d3) * p(apple|d3) = \left(\frac{1+\frac{5}{28}1000}{7+1000}\right) * \left(\frac{0+\frac{2}{28}1000}{7+1000}\right) = 0.01264$
- $p(q|d4) = p(orange|d4) * p(apple|d4) = \left(\frac{1+\frac{5}{28}1000}{9+1000}\right) * \left(\frac{1+\frac{2}{28}1000}{9+1000}\right) = 0.01277$

Relevance Model Example

- Construct a relevance model using the 2 most highly ranked documents

$$P(w|R) \approx \sum_{d \in D_{init}} P(w|M_d)P(M_d|q)$$

$$P(M_d|q) = \frac{P(q|M_d)P(M_d)}{\sum_{d \in D_{init}} P(M_d)P(q|M_d)}$$

  - $P(clementine|R) = \frac{1}{6} * \frac{0.0129}{0.0129+0.01277} = 0.0838$
  - $P(fruit|R) = \frac{1}{9} * \frac{0.01277}{0.0129+0.01277} = 0.055$
  - $P(apple|R) = \frac{1}{6} * \frac{0.0129}{0.0129+0.01277} + \frac{1}{9} * \frac{0.01277}{0.0129+0.01277} = 0.139$
  - …

- Documents are ranked according to the similarity of their model to that of the query

Clipping the Query Model

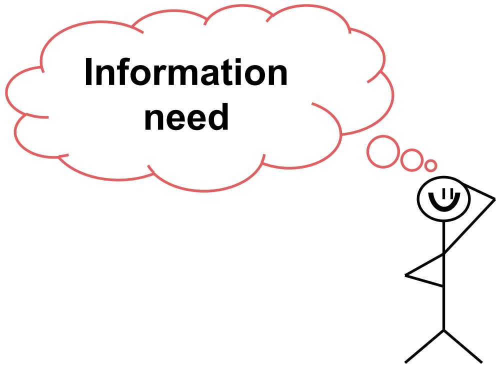
- The expanded query contains many terms
  - The amount of unique terms in documents provided for feedback
- Problems:
  - Query drift – why?
  - Computationally demanding
- Clipping the query model
  - Use the top v terms assigned the highest probability
  - Underlying assumption: these terms are most likely to represent the information need
  - $p(w|RM3) \stackrel{\text{def}}{=} \beta p_{MLE}(w|q) + (1-\beta)p(w|RM1^{clipped})$

Clipping - Example

- q_0 = “nobel prize”
- INEX collection
  - # of documents = 2,666,190
  - # of relevant documents for q_0 = 22
- Construct RM3 model
  - 3 relevant documents serve as true feedback
  - β = 0.2
  - Clipping: υ = 100
- statistics:
  - 79 out of 100 terms are stopwords!
  - The top 12 terms with the highest probability are:
  - nobel, prize, the, of, in, and, for, on, 2007, or, medicine, physiology

Additional Thoughts

- Given examples of relevant documents provided by the user
  - Will the expanded query always be closer to the centroid of the relevant documents?
- Suggest an approach for ameliorating this drawback
- How would you estimate $P(M_d \mid q)$?
- Using KL for ranking results in an IDF effect while using the MLE for the query language model
  - Rank equivalent to the query likelihood approach
- Does the IDF effect hold for any query model?
  - Raiber and Kurland 2017



Does the IDF Effect Hold for Any Query Model?

- M_d: Smoothed unigram document language model
- $M_d \stackrel{\text{def}}{=} (1-\lambda)M_d^{MLE} + \lambda\, M_C^{MLE}$
- For $w \epsilon d$:
  - $p_s(w|M_d)$ =$(1-\lambda)\; p(w|\; M_d^{MLE})$ + $\lambda\; p(w|\; M_C^{MLE})$
- For $w \notin d$:
  - $p_u(w|M_d)$ =$\lambda\; p(w|\; M_C^{MLE})$
- $-CE\left(M_q, M_d\right) = \sum_w p\left(w\middle|M_q\right) logp(w|M_d) \; =$

$$\sum_{w \in d} p\left(w\middle|M_q\right) logp_s(w|M_d) + \sum_{w \notin d} p\left(w\middle|M_q\right) logp_u(w|M_d) \; =$$

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  $\sum_{w \in d} p(w|M_q) log p_s(w|M_d) + \sum_{w \notin d} p(w|M_q) log p_u(w|M_d) =$
  $\sum_{w \in d} p(w|M_q) log \frac{p_s(w|M_d)}{p_u(w|M_d)} + \sum_{w} p(w|M_q) log p_u(w|M_d) =$
  $\sum_{w \in d} p(w|M_q) log \frac{p_s(w|M_d)}{\lambda\, p(w|\ M_C^{MLE})} + log\lambda - CE(M_q, M_C^{MLE}) \propto$

$$\sum_{w \in d} p(w|M_q) \log\left(1 + \frac{1-\lambda}{\lambda} \frac{p(w|\ M_d^{MLE})}{p(w|\ M_C^{MLE})}\right) + log\lambda$$